Module - 2

① Explain feature selection compare filter and wrapper methods

Feature selection:- Feature selection is the process of selecting the most important and relevant features (input variable) from a data set and removing unnecessary or irrelevant features

Why Feature selection is needed:-

⇒ Reduces the number of feature
⇒ Remove irrelevant and redundant data.
⇒ Improve model performance
⇒ Reduce training time
⇒ Helps prevent overfitting

1. Filter method:-

⇒ Feature are selected based on their statistical relationship with the target variables
⇒ The machine learning model is not required during selection
⇒ It is suitable for datasets with large number of features.

Ex:- ⇒ correlation
⇒ chi-square test
⇒ ANOVA
⇒ Information Gain

2. Wrapper method:-

⇒ Different combinations of feature are given to a machine learning algorithm.
⇒ The combination that give the best model performance is selected
⇒ It is more computationally expensive

Ex:- ⇒ forward selection
⇒ Backward Elimination
⇒ Recursive Feature Elimination (RFE)

(2) Describe the Decision

Decision Tree algorithm :-

A Decision Tree algorithm is a supervised machine learning algorithm used for classification and regression. It represents decisions in the form of tree structure.

Algorithm :-

⇒ Start with the complete dataset at the root node.
⇒ Select the best feature to split the data.
⇒ Calculate a splitting measure such as.
  * Information
  * Gini Index
  * Entropy.
⇒ Split the dataset into different branches based on the selected
⇒ Repeat the spliting process for each branch.
⇒ Stop splitting when:
  * All or most data belongs to the same class, or
  * No useful feature remain, or
  * A stopping condition is reached.
⇒ Assign the final class/value to the leaf node.
⇒ use the complete tree to make prediction for new dat

Advantages :-

⇒ Easy to understand and interpret.
⇒ Can be used for both classification and regression.
⇒ Require less data preprocessing.
⇒ Can handle numerical and categorical data
⇒ Does not require feature scaling or normalization.

(3) Explain Principal component Analysis (PCA) and its application.

Principal component Analysis :-

Is a dimensionality reduction technique used in machine learning and Data science. It reduces the number of feature in a dataset while Preserving as much important information as possible

steps involved in PCA

standardize the data :- Bring all features to a similar scale

Calculate the covariance matrix :- Identify relationship b/w features

Find eigenvalue and eigenvectors :- Determine the importance and direction of each component

Select Principal components :- choose the components that retain the most variance.

Transform the data :- convert the original data into a smaller set of features

Applications :-

1. Dimensionality Reduction :- Reduces the number of features in large datasets.

2. Image Processing :- Reduces image size and helps in facial recognition.

3. Data visualization :- converts high-dimensional data into 2D or 3D for easy visualization

4. Noise Reduction :- Removes less important variations from data

5. machine learning :- Reduce computation time can improve model performance.

6. Finance :- Helps analyze patterns in stock market data

7 Healthcare :- Helps analyze medical data and identify important patterns.